# Milindra Pratap Singh

**Senior Software Engineer** | **Backend Engineer** | **Distributed Systems** | **FDE / Customer Deployment**

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## PROFESSIONAL SUMMARY

Senior Software Engineer with **7+ years** of experience building large-scale distributed systems, cloud-native applications, AI-powered voice platforms, and enterprise software across **Vibrium AI, Fanatics, Microsoft, Mobile Premier League (MPL), and Samsung Research Institute**. Expertise in **Python, Java, C#, AWS, Microservices, LLMs, PSTN Telephony, STT/TTS,** and **customer/on-premise deployment**. Proven track record of leading engineering projects, mentoring teams, optimizing system performance, troubleshooting production systems, and deploying SaaS solutions into enterprise customer infrastructure through close collaboration with Business, Integration Engineering, DevOps, Infrastructure, and Security/VAPT teams.

## EXPERIENCE

### Vibrium AI
Oct 2025 – Present
*Senior Software Engineer*
Hyderabad, India

- Architected an AI-powered voice automation platform handling complete **PSTN call flows** with intelligent AI agents for inbound/outbound customer interactions.
- Led and mentored 3-4 engineers through technical design reviews, implementation, code quality, troubleshooting, and delivery.
- Built real-time voice pipelines integrating **STT, LLMs, and TTS** with interruption handling, VAD, echo cancellation, and noise suppression.
- Optimized **LLM token consumption and prompt orchestration**, reducing inference cost while maintaining response quality.
- Designed scalable **event-driven AWS microservices** for telephony orchestration, workflow management, campaign automation, webhooks, and enterprise integrations.
- **Led on-premise deployment of the SaaS AI voice solution into customer infrastructure**, adapting services, configuration, networking, security, observability, and deployment architecture to enterprise requirements.
- Worked directly with customer **Business, Integration Engineering, DevOps, Infrastructure, and VAPT/Security teams** to understand business requirements, technical dependencies, deployment constraints, and security requirements.
- Coordinated customer integrations involving **PBX/SIP, APIs, authentication, networking, databases, S3/cloud data exchange, webhooks, WhatsApp Business API, and external telephony/dialer systems**.
- Supported **VAPT remediation**, coordinating vulnerability findings and fixes across engineering, DevOps, and security teams.
- Troubleshot deployment and production issues by analyzing logs, tracing failures across services, identifying root causes, and coordinating resolution with relevant teams.
- Designed WebSocket scaling architecture using **NLB, Auto Scaling, Redis-backed stateless sessions, SQS, and authorization controls**.
- Worked on usage quota/credit models, workflow and campaign architecture, webhook storage, order synchronization, and enterprise deployment planning.
- **Tech:** Python, AWS, Docker, Redis, DynamoDB, Aurora/PostgreSQL, SQS, EventBridge, NLB, EC2, CloudWatch, S3, LiveKit, SIP, PSTN, LLMs, STT, TTS, WhatsApp Business API, CI/CD, HLD, LLD

### Fanatics — E-Commerce
Mar 2025 – Oct 2025
*Software Development Engineer III*
Hyderabad, India

- Managed end-to-end order lifecycle across Payment, Order, and Fraud Orchestrator microservices.
- Built a **scheduler service from scratch** to update and clean Elasticsearch fulfillment records via gRPC, achieving **~1,200 records/sec** with asynchronous Golang workers.
- Improved reliability and scalability of order-processing pipelines for high-volume global e-commerce traffic.
- **Tech:** Golang, Java, C#, AWS, Temporal, gRPC, Elasticsearch, SQL, CI/CD

### Microsoft IDC
Oct 2021 – Feb 2025
*Software Development Engineer*
Hyderabad, India

- Designed a **large-scale proxy layer** for a strategic platform migration initiative across the OneDrive and SharePoint ecosystem.
- Proxied **12,000+ APIs** across OneDrive, Photos, Xbox, OneNote, WXP, and third-party applications.
- Supported migration of **500M+ user accounts** with minimal customer impact across Photos, Xbox, and SWAY teams.
- Resolved critical production incidents during migration cycles, improving platform reliability.
- **Tech:** C#, SQL, Distributed Systems, OOP, LLD, HLD, Microservices

## TECHNICAL SKILLS

**Languages**

Python | Golang | Java | C# | C/C++

**Cloud & Infrastructure**

AWS | Docker | CI/CD | Redis | EC2 | Lambda | ECR | DynamoDB | S3 | SQS | EventBridge | NLB | CloudWatch | Aurora

**Architecture**

Microservices | Distributed Systems | Event-Driven | LLD | HLD | gRPC | Temporal | WebSockets

**AI / Voice / Telephony**

LLM | AI Voice Agents | PSTN | STT | TTS | VAD | Prompt Engineering | SIP | LiveKit | Telephony

**Data & Algorithms**

SQL | PostgreSQL | MySQL | Redis | DynamoDB | Elasticsearch | DSA | DBMS

## CUSTOMER / ENTERPRISE DEPLOYMENT

**On-Premise SaaS Deployment**
Customer infrastructure deployment, environment configuration, networking, authentication, observability, deployment planning, production support.

**Cross-Functional Collaboration**
Business stakeholders, Integration Engineering, DevOps, Infrastructure, Security/VAPT, Product Engineering, and customer technical teams.

**Enterprise Integrations**
PBX/SIP, telephony/dialer, APIs, webhooks, WhatsApp Business API, databases, S3/cloud data exchange, networking and security.

## EDUCATION

### National Institute of Technology, Hamirpur
Dual Degree — B.Tech + M.Tech
Computer Science & Engineering
July 2014 – June 2019

## PUBLICATION

### A Tool Prototype for Privacy Preserving Distributed Data Mining and Analytics
**Milindra Pratap Singh**, Sudhanshu Raman, Swapnil Shrivastava, Supriya N. Pal, N. Sarat Chandra Babu
*CiiT International Journal*

**Mobile Premier League (MPL)** Apr 2021 – Oct 2021
*Software Development Engineer* Bangalore, India

- Built customer-facing dashboard services for wallet, gameplay history, banking information, and profiles.
- Developed support ticketing with real-time customer–agent communication.
- Built matchmaking notification services for game-ready alerts.
- **Tech:** Java, AWS, gRPC, Redis, SQL, Microservices

**Samsung Research Institute** Aug 2019 – Apr 2021
*Software Engineer* Noida, India

- Worked on **IMS framework, SIP protocol, LTE, VoLTE, and ViLTE** for Samsung smartphones with global operators.
- Managed VoWiFi services for Galaxy S20 and S21 across South-West Asia operators.
- Built automated log-analysis tools that accelerated production issue diagnosis.
- **Tech:** Java, C, C++, SIP, LTE, IMS

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